MAE 3270

Final Course Project
Improved Design Analysis
Term: Fall 2025
Yussef Oufkir (yro2)

1. **CAD Model with relevant dimensions.**

Figure 1: Rendering of the wrench's CAD Model

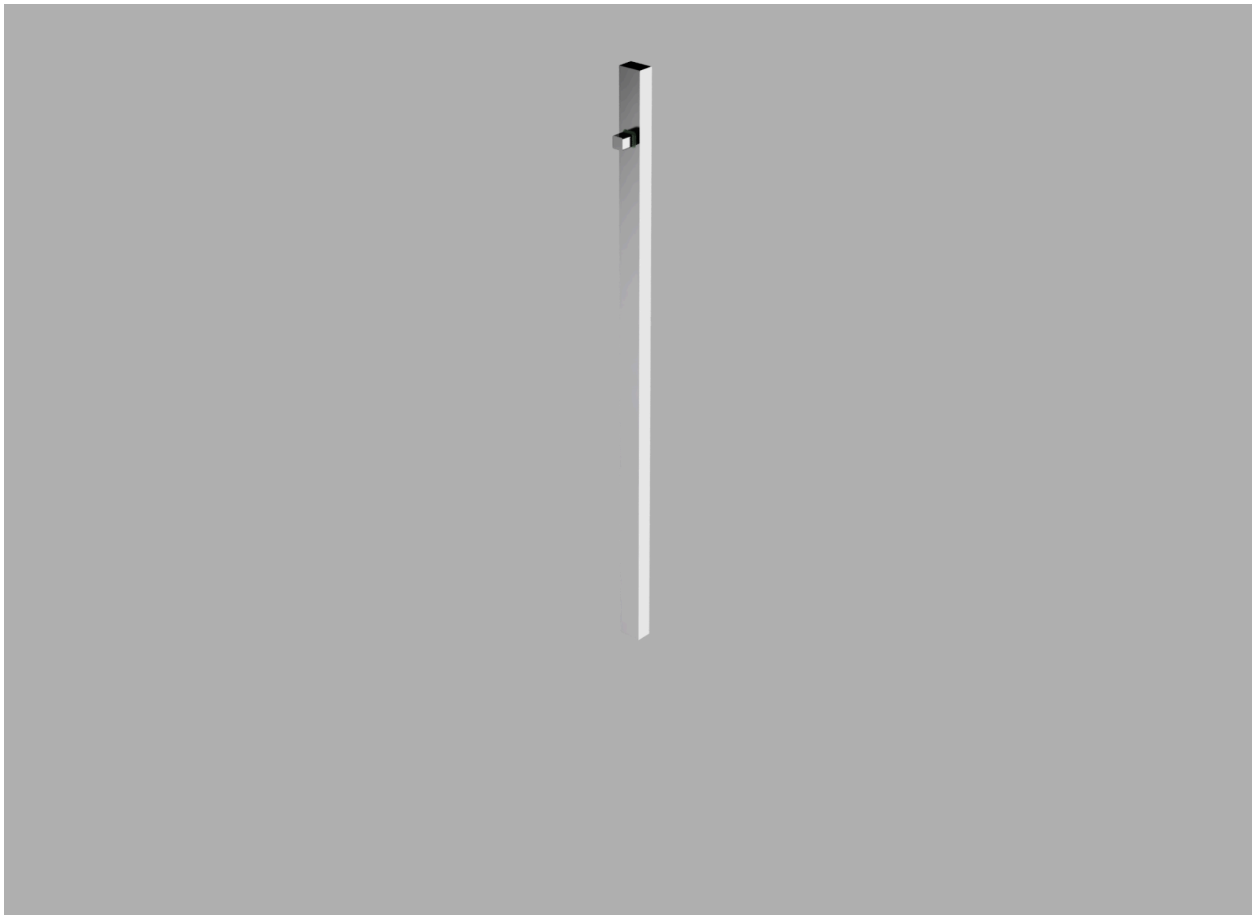


Figure 2: Image showing the thickness b = 0.5 in of the wrench.

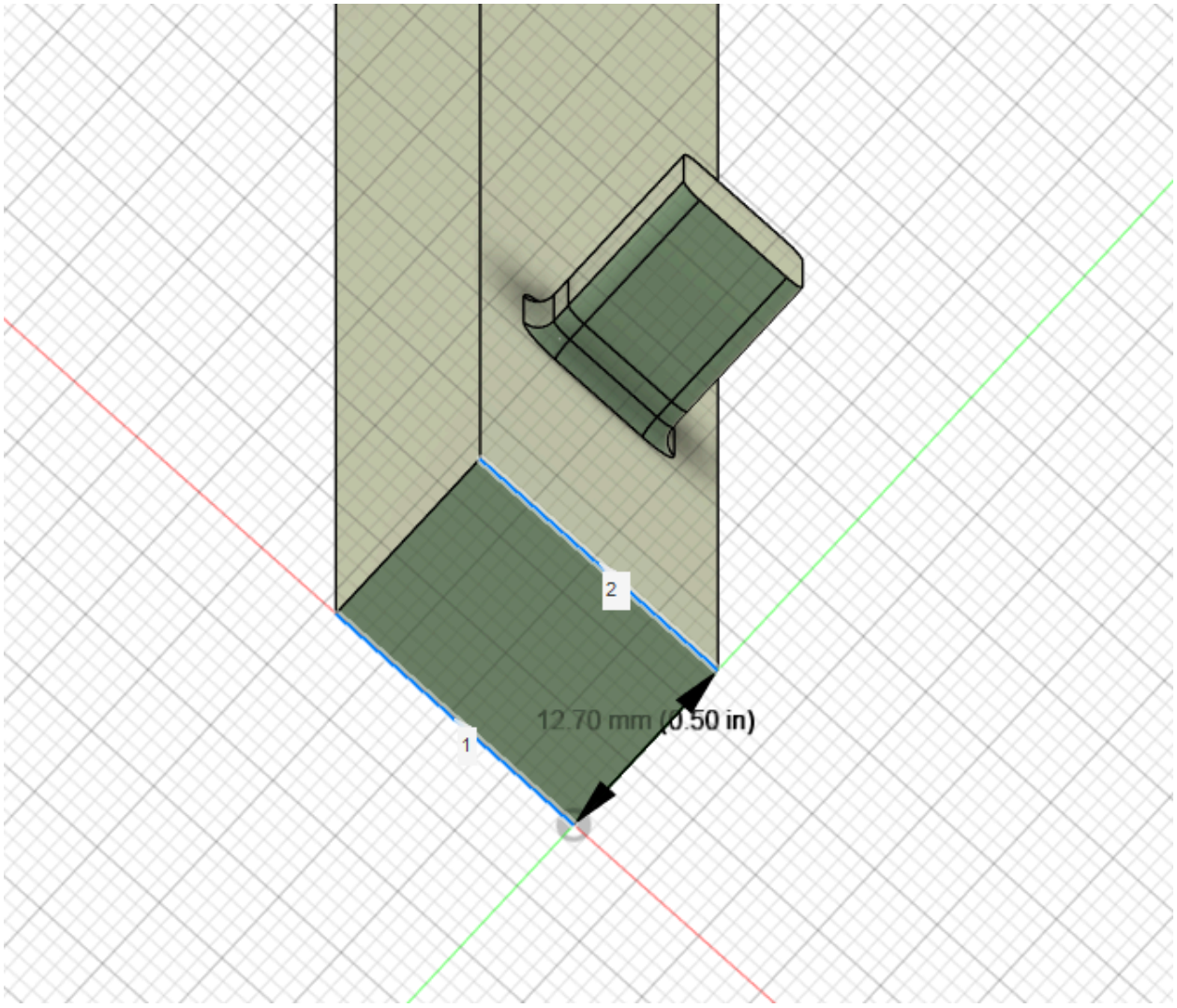


Figure 3: Image showing the width h = 0.75 in of the wrench.

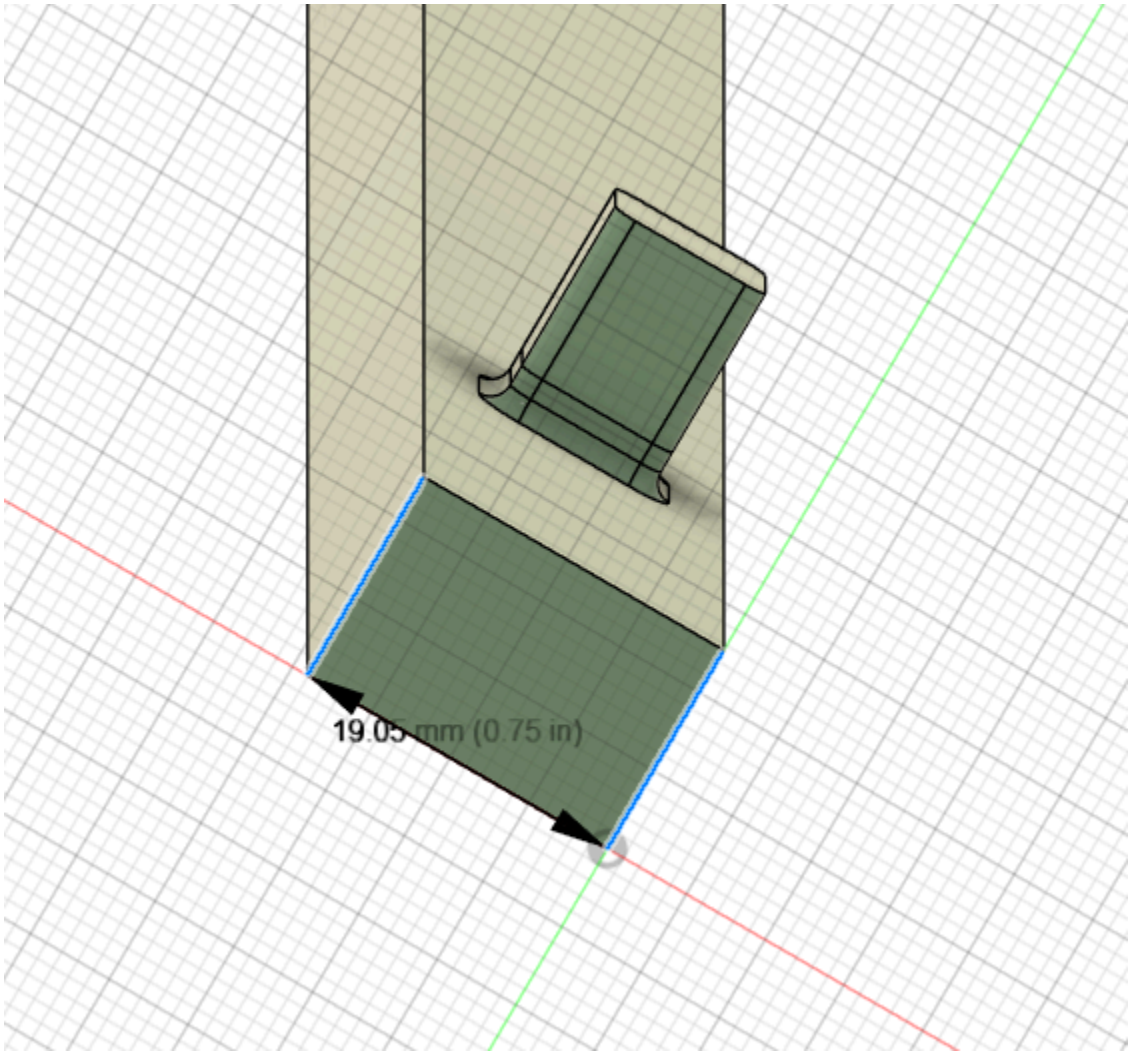


Figure 4: Image showing the filet of 0.05 in between the drive and the shaft.

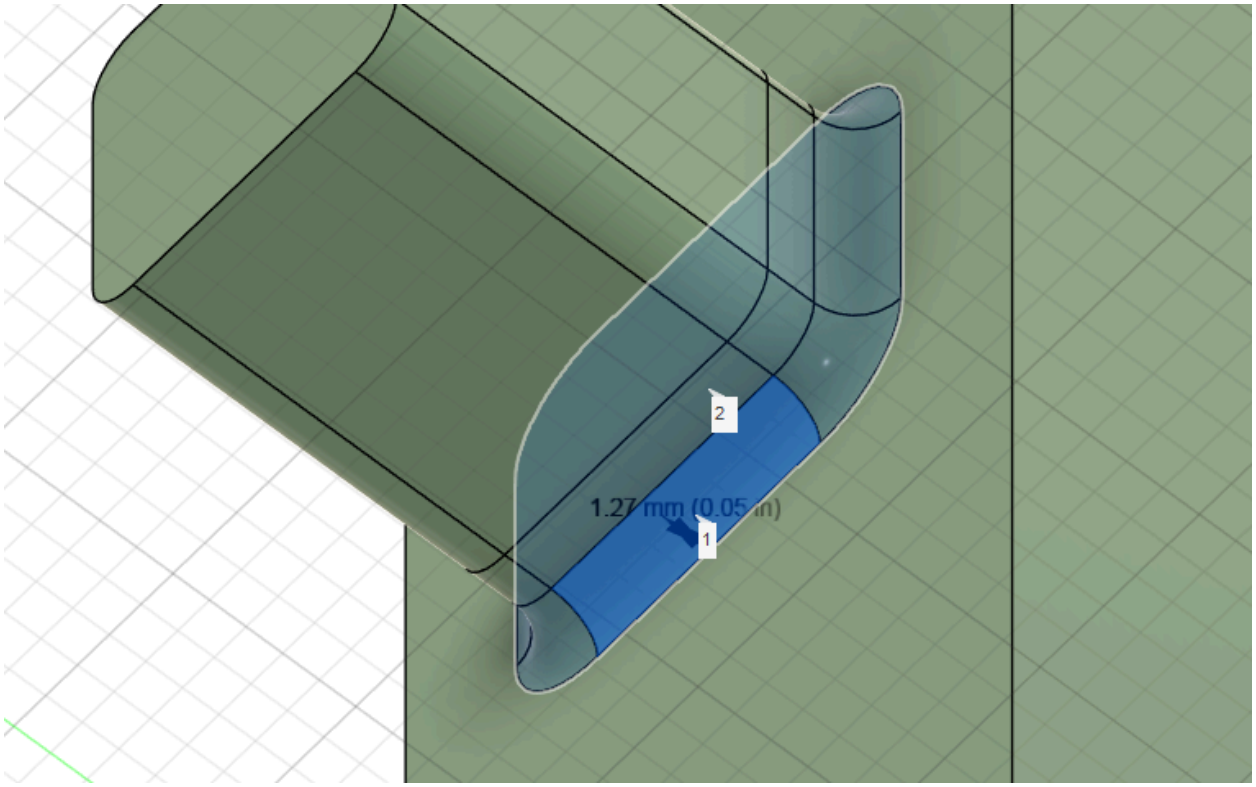
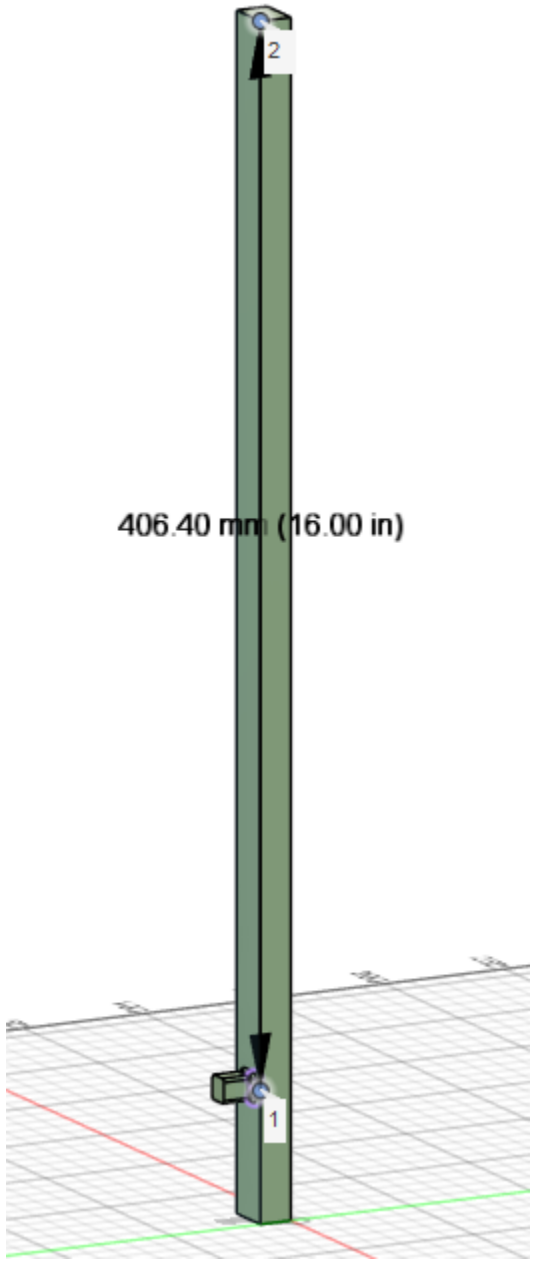


Figure 5: Image showing the length from the end of the shaft to the drive, L = 16 in.



2. The Material Used

The material used in the final design is aluminum 7075-T6, which is a premium, high-strength aluminum alloy belonging to the 7000-series, which is alloyed primarily with zinc. It is highly regarded for its exceptional strength-to-weight ratio. The T6 temper indicates it has been solution heat-treated and aged to achieve its maximum strength and hardness. Due to these properties, 7075-T6 is widely used in demanding aerospace applications, such as aircraft structural components.

The relevant material properties were extracted from GRANTA and include:

- Young's Modulus: $E = 10.5 * 10^6 psi$
- Poisson's Ratio: $\nu = 0.33$
- Tensile strength: $S_u = 80 * 10^3 psi$
- Fatigue strength: $S_{fatigue} = 23 * 10^3 psi$
- Fracture toughness: $K_{IC} = 24.3 * 10^3 psi\sqrt{in}$

3. FEM Model Conditions

Figure 6: Diagram indicating the fixed boundary condition at the drive, 0 displacement in all directions.

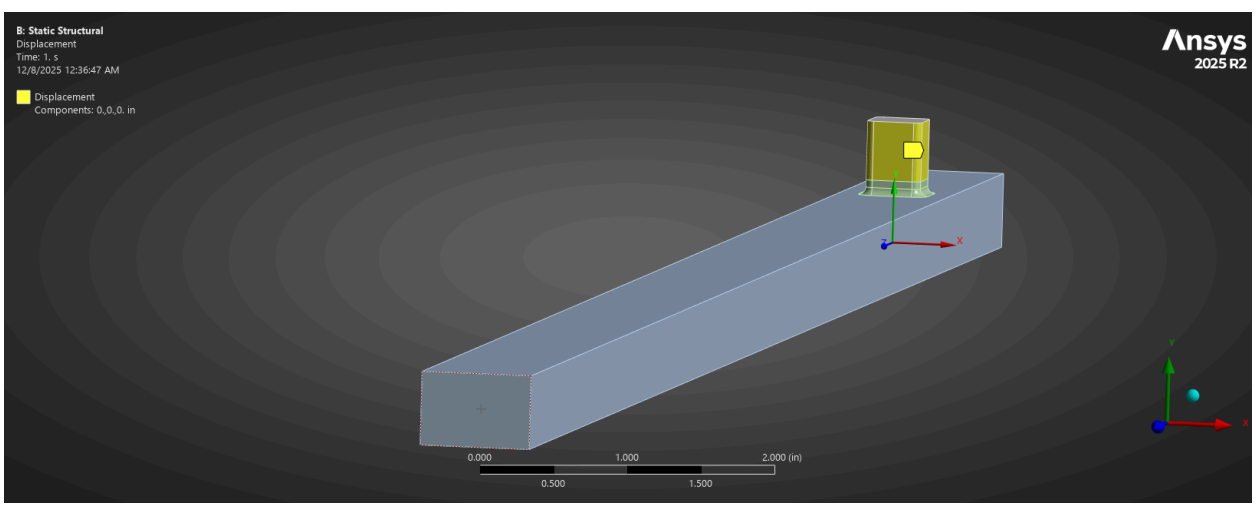
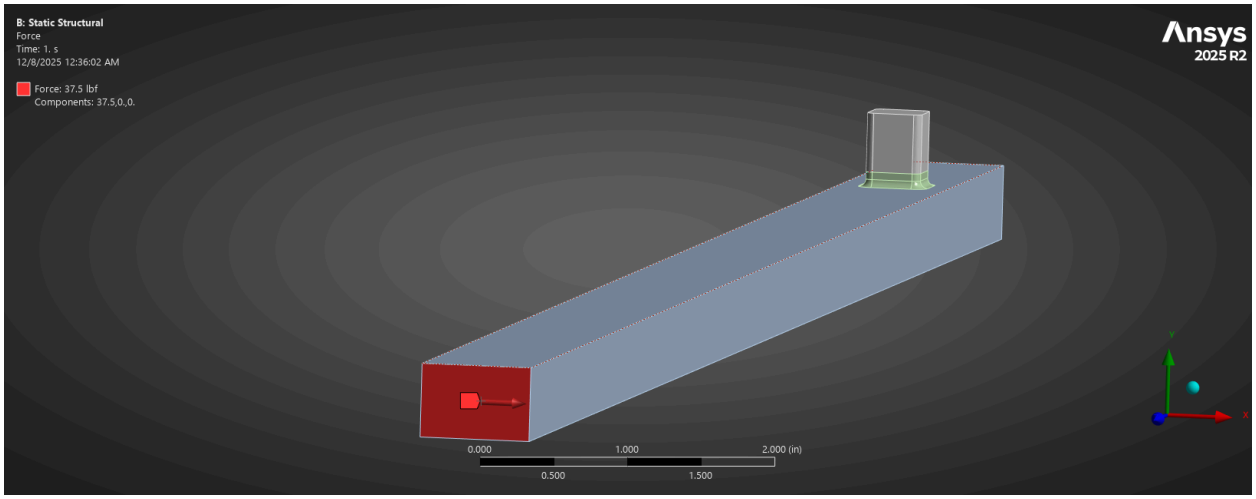
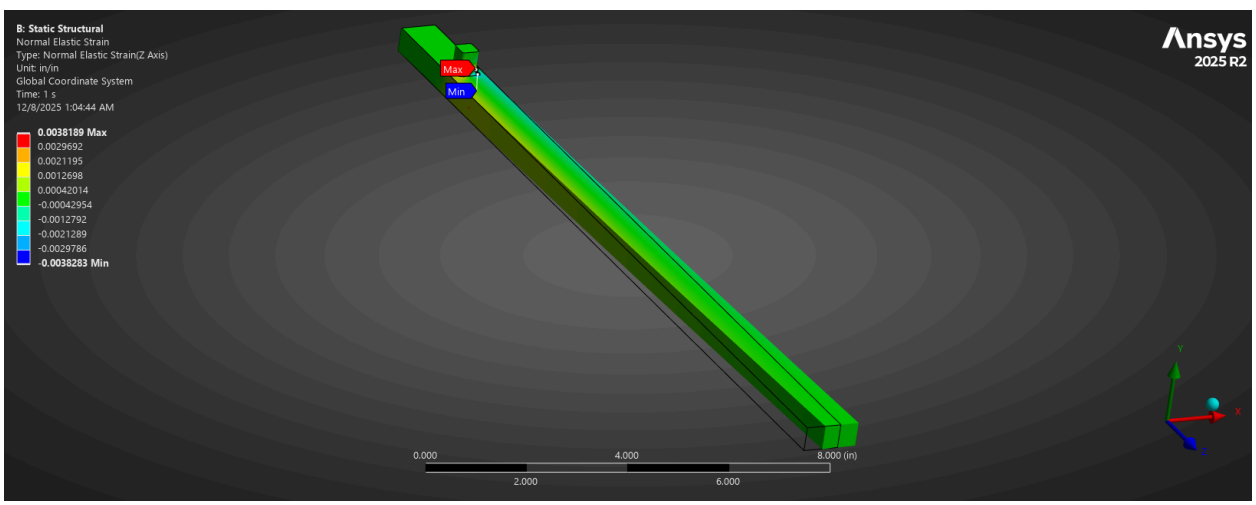


Figure 7: Diagram indicating the force applied at the load point, 37.5 lbf in the x-direction.



4. Normal Strain Contours

Figure 8: Image indicating the normal strain contours in the strain gauge direction (z axis).



5. Maximum Principal Stresses

Figure 9: Image indicating the maximum principal stress in the model.

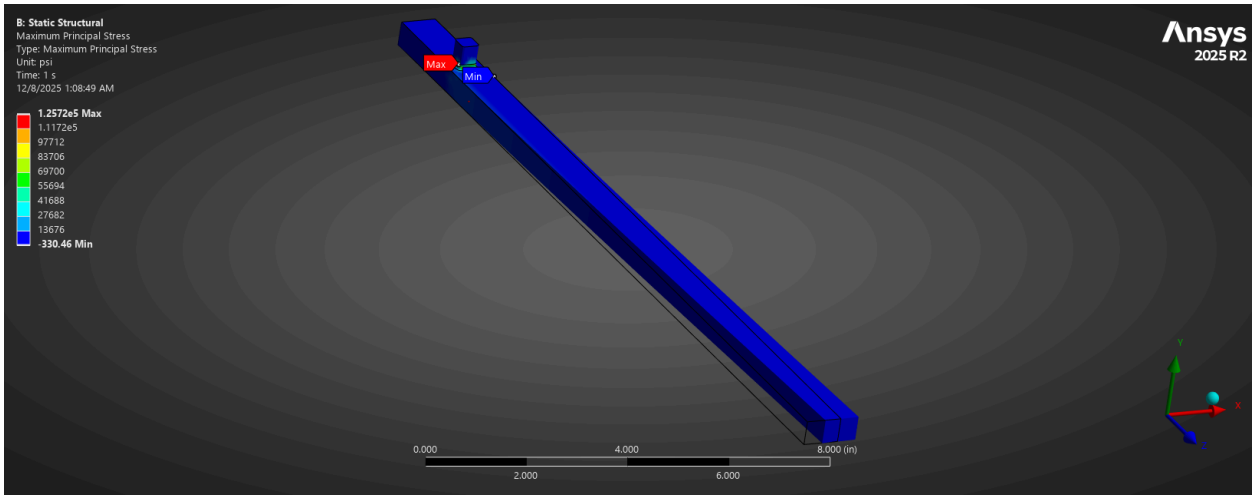
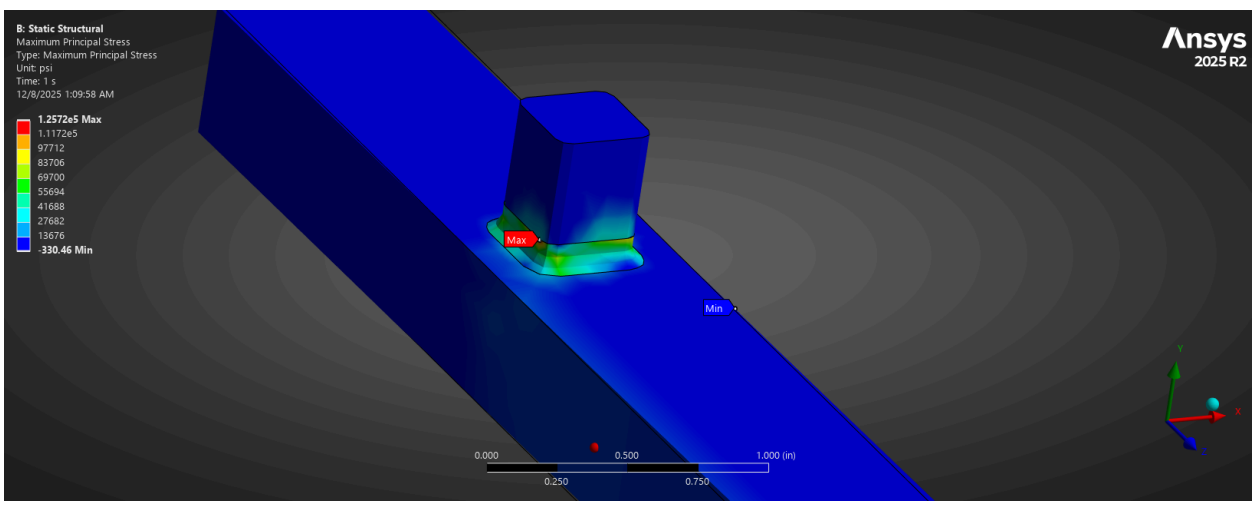


Figure 10: Image indicating the maximum principal stress in the model near the drive.



6. Results from FEM Calculation

Figure 11: Image indicating the normal stress in the model.

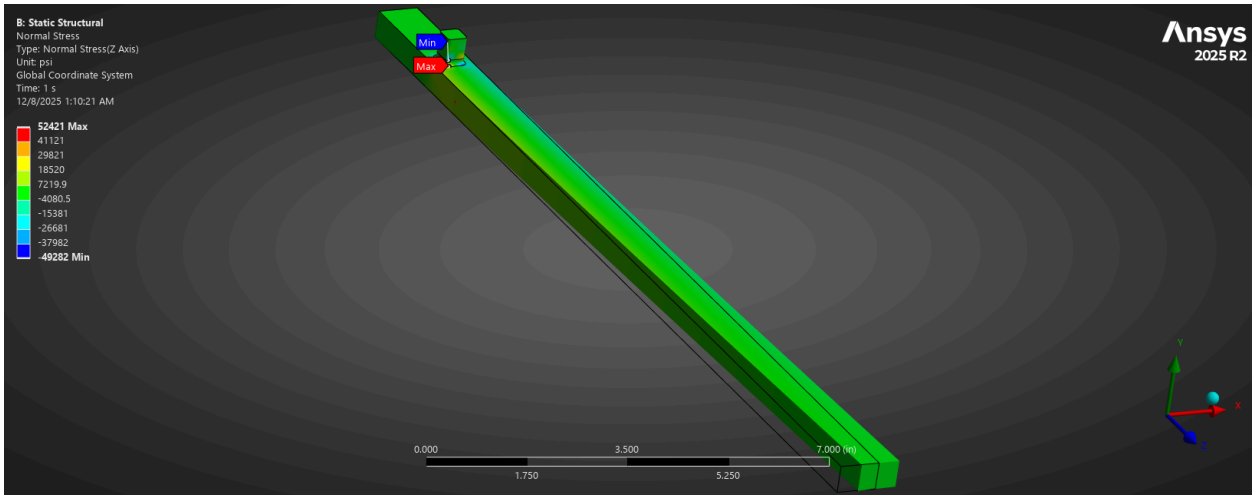
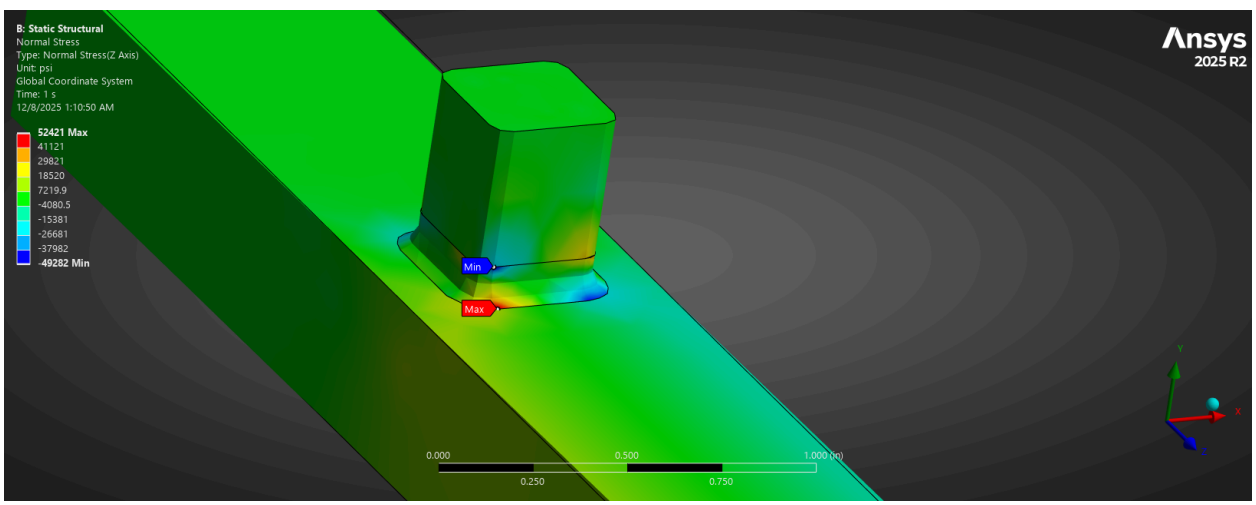
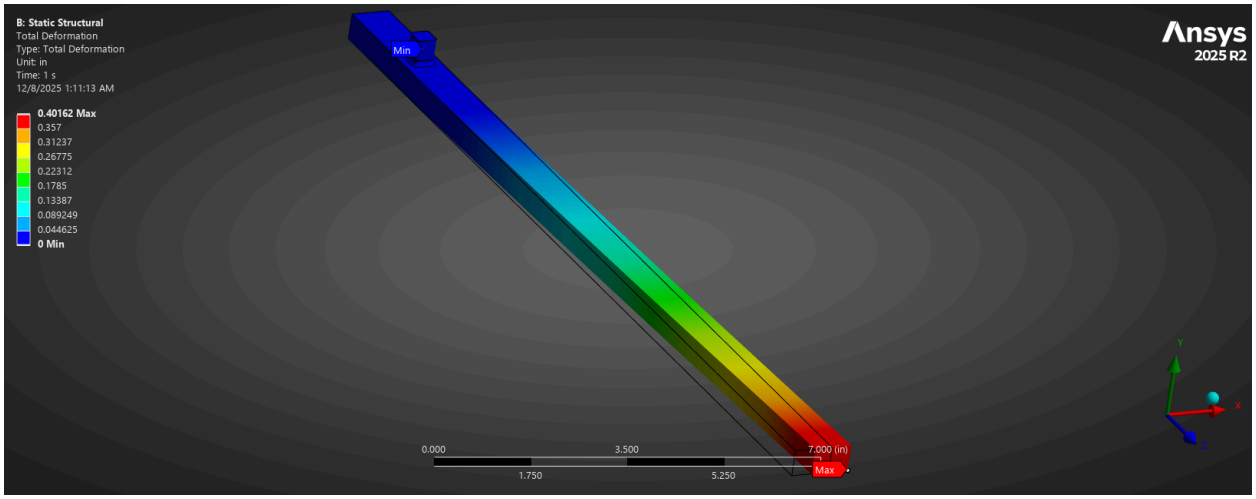


Figure 12: Image indicating the normal stress in the model near the drive.



Even by mitigating the stress concentration by introducing a filet between the drive and the wrench's shaft, the maximum normal stress still occurs in this region with a value of 52.4 ksi.

Figure 13: Image indicating the total deflection.



The load point deflection in this model is 0.40162 in.

Figure 14: Image indicating the location of the strain probe, 1 inch away from the drive.

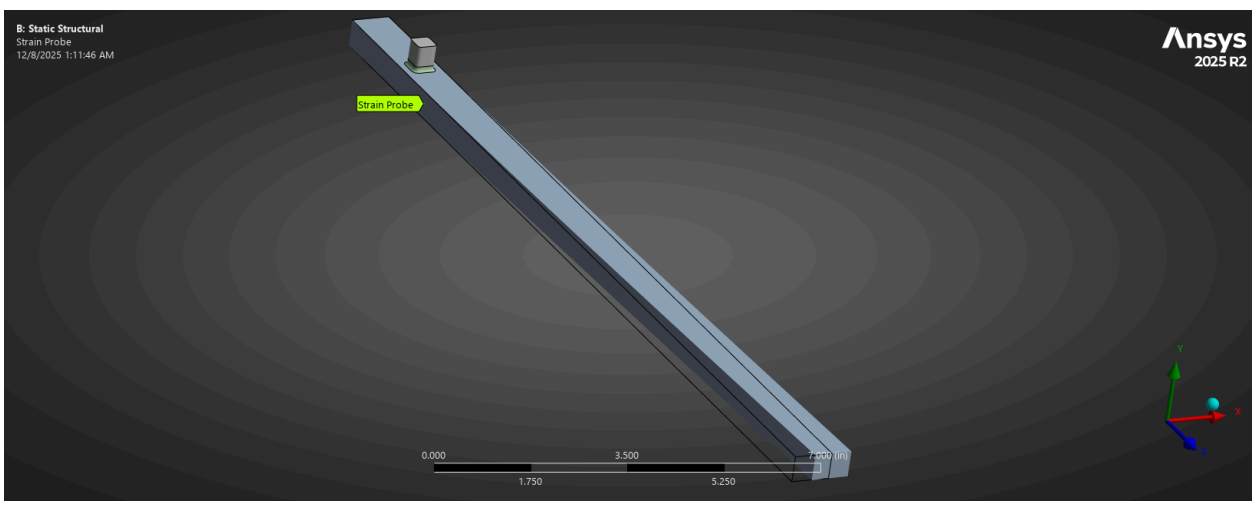


Figure 14: Table indicating the strain recorded at the strain gauge location. The relevant value is that in the z direction.

Maximum Value Over Time	
Normal - X Axis	-3.7767e-004 in/in
Normal - Y Axis	-3.7725e-004 in/in
Normal - Z Axis	1.1433e-003 in/in

7. Torque Wrench Sensitivity

For our strain gauge setup, the torque wrench sensitivity in mV/V can be calculated as:

$$Sensitivity = \varepsilon_{gauge} * 1000 = 1.14 * 10^{-3} * 1000 = 1.14\, mv/V$$

8. Strain Gauge Setup

The setup used is a half-bridge strain gauge setup, which uses two active strain gauges within a Wheatstone bridge circuit. The strain gauge is located at a distance c = 1 in from the drive. The strain gauge selected for the torque wrench design is the Micro-Measurements CEA-XX-250UWA-120, a general-purpose linear pattern gauge. This uniaxial gauge has a nominal resistance of 120 Ω and a gauge length of 0.250 inches. Crucially for the design's physical feasibility, the overall package dimensions are 0.450 inches in length and 0.180 inches in width. This small size allows the gauge to be easily bonded to the wrench handle's surface, which is 0.75 in wide and 0.5 in thick.

Hand Calculations confirming Validity of Design:

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--- Analysis: 3. Optimized 7075-T6 Aluminum ---
Dimensions: h=0.75 in, b=0.5 in
----------------------------------------
Moment Max (in-lbf): 600.00
Moment @ Gauge (in-lbf): 562.50
----------------------------------------
Max Stress (ksi): 12.80
Strain @ Gauge (ustrain): 1143
Deflection (in): 0.2774
----------------------------------------
Output (mV/V): 1.143
Static SF (Xo=4.0): 6.25 | PASS
Fatigue SF (Xs=1.5): 1.80 | PASS
Fracture SF (Xk=2.0): 4.72 | PASS
----------------------------------------
Design **MEETS ALL REQUIREMENTS**
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